

Math 252Z - Calculus II - Final Exam

December 9, 2025

Last Name: _____, First Name: _____

1. There are 17 problems. Some problems have multiple parts. The total credit is 100.
2. No collaboration is allowed for this exam. The work on this exam is to be yours and yours alone. Failure to adhere to this policy will result in a zero on this exam. Any instances of suspected cheating will also be reported to the Dean of Students' Office.
3. No calculators, books or other material are permitted.
4. In order to receive full credit you need to show your work, and present your answers as clearly as possible. Final answers without work may not receive any credit, unless a problem is a fill-in-blank or multiple choice problem.
5. Answers must be exact (like $\sqrt{2}$), not approximate (like 1.414), unless the problem specifically indicates otherwise.
6. Simplify where appropriate. Quantities like $\sqrt{9}$, $\ln e$, inverse trigonometric functions at special values, or trigonometric functions at special angles should be calculated. Reduce all fractions. Answers that are not simplified may result in loss of points.
7. If an answer line is provided for the answer, place your final answer in it. Otherwise, circle, box, or underline each final answer as appropriate for clarity.
8. Failure to follow directions specific to a problem will result in loss of points.
9. If you work a problem in two different ways, please clearly indicate which one you want us to grade. If you use two methods, one of which is wrong, and neither method is crossed out, you will not receive full credit.
10. Put away and **silence** your cell phones, tablets, laptops, etc.
11. Once this exam begins, you will have 120 minutes to complete your solutions.

1. [2 pts] Set up the partial fraction decomposition of the following rational function. DO NOT determine the numerical values of the coefficients. DO NOT integrate the function.

$$\frac{x^3 - x^2 + 3x + 4}{x(x+2)^3(x^2+1)} = \underline{\hspace{10cm}}$$

2. [2 pts] Complete the following arguments of using Comparison Test to determine if an improper integral is convergent or divergent.

The improper integral $\int_1^{\infty} \frac{\cos^2 x}{\sqrt[7]{x^5 + 1}} dx$ is _____ (choose “convergent” or “divergent” to fill in the blank) because $\frac{\cos^2 x}{\sqrt[7]{x^5 + 1}}$ _____ (choose one between A and B , and then choose one among C , D , E and F).

A). $\geq$ B). $\leq$, C). $\frac{1}{x^{1/5}}$, D). $\frac{1}{x^{1/7}}$, E). $\frac{1}{x^{5/7}}$, F). $\frac{1}{x^{7/5}}$

3. [5 pts] Evaluate $\int \frac{x^6 + x^{8/3}}{x^4} dx$.

4. [7 pts] 1). Simplify the sum $R_n = \sum_{i=1}^n \left(\frac{10i}{n} - \frac{12i^2}{n^2} \right) \cdot \frac{2}{n}$. Write your answer as a function of n without any summation signs.

- 2). Evaluate $\lim_{n \rightarrow \infty} \sum_{i=1}^n \left(\frac{10i}{n} - \frac{12i^2}{n^2} \right) \cdot \frac{2}{n}$ by using your result from Part 1).

- 3). Express $\lim_{n \rightarrow \infty} \sum_{i=1}^n \left(\frac{10i}{n} - \frac{12i^2}{n^2} \right) \cdot \frac{2}{n}$ as a definite integral.

Answer: $\lim_{n \rightarrow \infty} \sum_{i=1}^n \left(\frac{10i}{n} - \frac{12i^2}{n^2} \right) \cdot \frac{2}{n} = \int_{\underline{\hspace{1cm}}}^{\overline{\hspace{1cm}}} \underline{\hspace{1cm}} \hspace{1cm} dx.$

5. [5 pts] Evaluate $\int x^5 e^{-2x^6} dx$.

6. [5 pts] Evaluate $\int \cos(9x) \cos(4x) dx$.

7. [6 pts] Evaluate $\int \sqrt{x} \ln x dx$.

8. [7 pts] Evaluate $\int e^x \cos(3x) dx$.

9. [6 pts] Evaluate $\int \frac{5}{(2x-1)(x+2)} dx$.

10. [7 pts] Evaluate $\int \frac{1}{x^2 \sqrt{x^2 + 4}} dx$.

Note: Simplify your answer so that it does not contain any trigonometric composed with inverse trigonometric terms.

11. [7 pts] Evaluate the improper integral $\int_1^{\infty} \frac{(\arctan x)^2}{1 + x^2} dx$ or show that it is divergent.

Note: Use the limit definition of improper integrals to compute the above integral. Do not use Comparison Test.

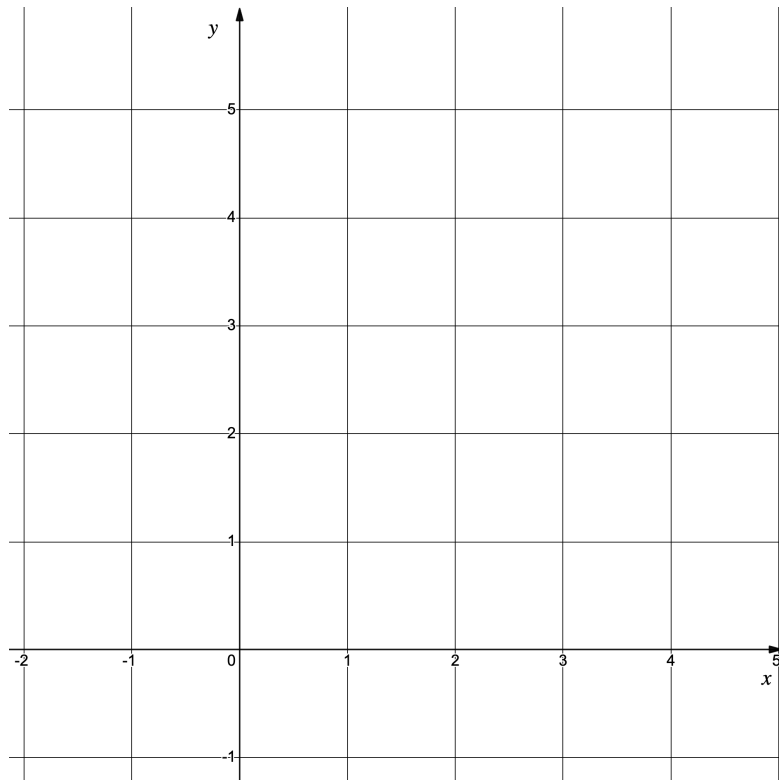
12. [7 pts] Find the average value of $f(x) = \sin^3 x \cos^3 x$ on the interval $[0, \frac{\pi}{2}]$.

13. [7 pts] A thin metal bar with a density function of $\rho(x) = 4 + \frac{16}{x^3}$ (in kg/m) lies on the x -axis between $x = 1$ and $x = 2$ (in meters).

1). Find the mass of the bar. 2). Find the center of mass of the bar.

Answer: 1). The mass $m =$ _____ kg. 2). The center of mass $\bar{x} =$ _____ m.

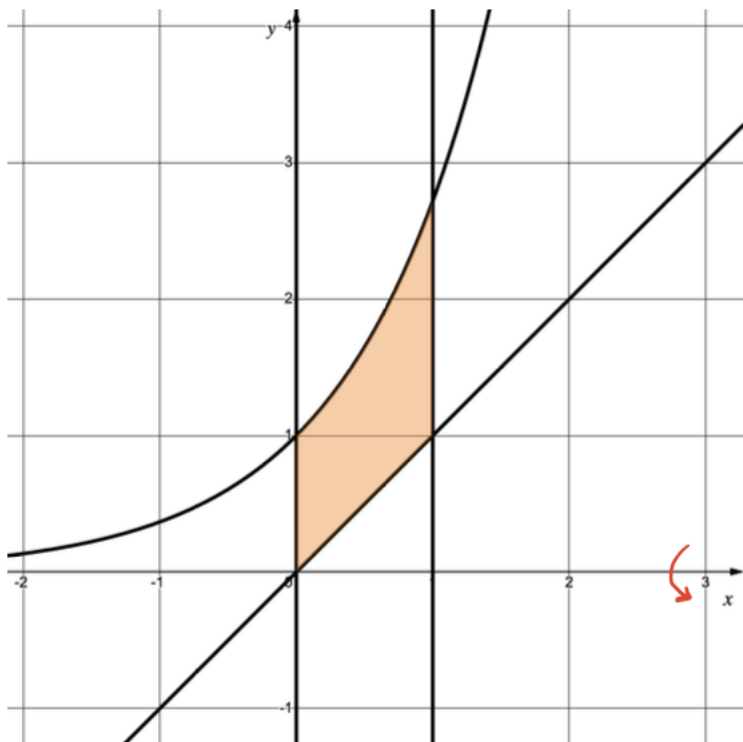
14. [7 pts] 1). Sketch the graph of the region **in the first quadrant** bounded by $y = 3 - x^2$, $y = 2x$ and **the y-axis**. *Make sure to shade the region.*



- 2). Find the **area** of the region in Part 1).

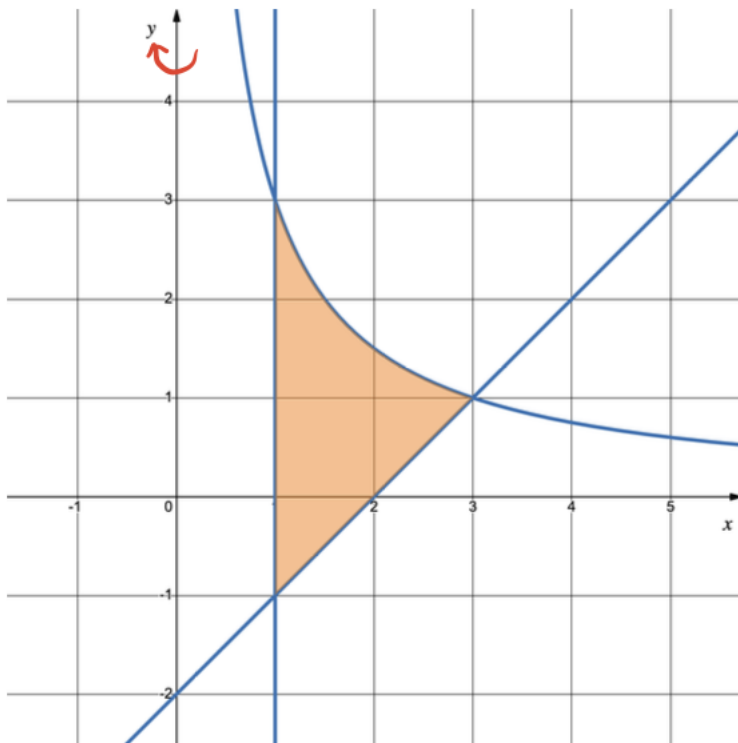
Answer: The **area** of the region is _____.

15. [7 pts] The following shaded region is bounded by $y = e^x$, $y = x$, the y -axis, and $x = 1$. Find the **volume** of the solid obtained by rotating this region about x -axis.



Answer: The **volume** of the solid is _____.

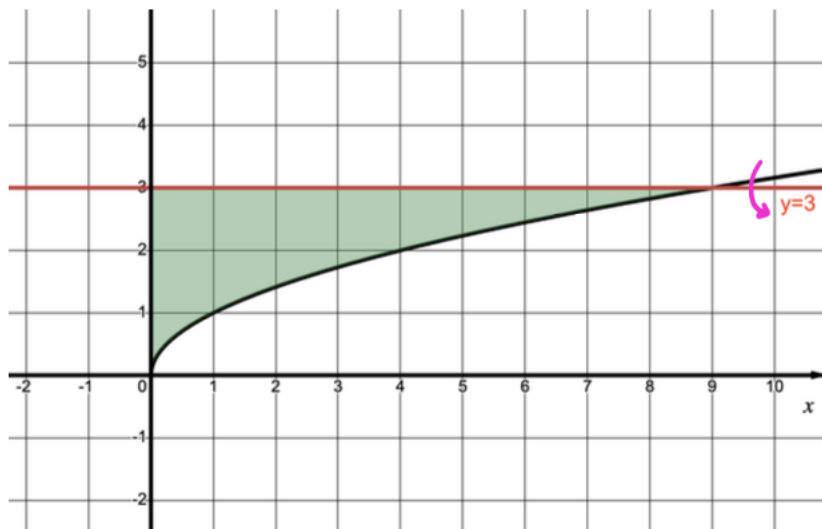
16. [7 pts] The following shaded region is bounded by $y = \frac{3}{x}$, $y = x - 2$, and $x = 1$ in the first quadrant. Find the **volume** of the solid obtained by rotating this region about y-axis.



Answer: The **volume** of the solid is _____.

17. [6 pts] The the following shaded region is bounded by $y = \sqrt{x}$, $y = 3$ and the y -axis.

Set up, **but do not evaluate**, integrals to compute of the volume of the solid obtained by rotating this region about the horizontal line $y = 3$ by using both disk/washer and cylindrical shell methods.



- 1). Disk/Washer Method:

$$\text{Volume of the Solid} = \int_{\underline{\hspace{1cm}}}^{\underline{\hspace{1cm}}} \underline{\hspace{10cm}} d\underline{\hspace{1cm}}$$

- 2). Cylindrical Shell Method:

$$\text{Volume of the Solid} = \int_{\underline{\hspace{1cm}}}^{\underline{\hspace{1cm}}} \underline{\hspace{10cm}} d\underline{\hspace{1cm}}$$

Note: You do not need to simplify the integrand functions or evaluate the above integrals.

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Formula Sheet

$$\sum_{i=1}^n c = cn, \quad \sum_{i=1}^n i = \frac{n(n+1)}{2}$$

$$\sum_{i=1}^n i^2 = \frac{n(n+1)(2n+1)}{6} = \frac{2n^3 + 3n^2 + n}{6}$$

$$\sum_{i=1}^n i^3 = \left[\frac{n(n+1)}{2} \right]^2 = \frac{n^4 + 2n^3 + n^2}{4}$$

$$\sin^2 x + \cos^2 x = 1, \quad \sin^2 x = 1 - \cos^2 x, \quad \cos^2 x = 1 - \sin^2 x$$

$$\sec^2 x = \tan^2 x + 1, \quad \tan^2 x = \sec^2 x - 1$$

$$\sin^2 x = \frac{1 - \cos 2x}{2}, \quad \cos^2 x = \frac{1 + \cos 2x}{2}, \quad \sin 2x = 2 \sin x \cos x$$

$$\sin A \cos B = \frac{1}{2} [\sin(A - B) + \sin(A + B)]$$

$$\cos A \cos B = \frac{1}{2} [\cos(A + B) + \cos(A - B)]$$

$$\sin A \sin B = \frac{1}{2} [\cos(A - B) - \cos(A + B)]$$

$$\int \cot x \csc x \, dx = -\csc x + C, \quad \int \csc^2 x \, dx = -\cot x + C$$

$$\int \tan x \, dx = \ln |\sec x| + C, \quad \int \cot x \, dx = \ln |\sin x| + C$$

$$\int \sec x \, dx = \ln |\sec x + \tan x| + C, \quad \int \csc x \, dx = \ln |\csc x - \cot x| + C$$

$$\int \frac{1}{a^2 + x^2} = \frac{1}{a} \arctan\left(\frac{x}{a}\right) + C, \quad \int \frac{1}{\sqrt{a^2 - x^2}} = \arcsin\left(\frac{x}{a}\right) + C$$

$$\int_1^\infty \frac{1}{x^p} \, dx = \begin{cases} \text{is convergent} & \text{if } p > 1 \\ \text{is divergent} & \text{if } 0 < p \leq 1. \end{cases}$$

$$\int_0^1 \frac{1}{x^p} \, dx = \begin{cases} \text{is convergent} & \text{if } 0 < p < 1 \\ \text{is divergent} & \text{if } p \geq 1. \end{cases}$$

$$m = \int_a^b \rho(x) \, dx, \quad \bar{x} = \frac{\int_a^b x \rho(x) \, dx}{\int_a^b \rho(x) \, dx}$$